

Large TVs

Cinema is obsolete. Bring the movies home, but buy one of these first

The visual revolution has brought about a thought revolution too. Larger screens were always available, but consumers passed up 29-inch CRTs for the *de facto* 21-inch CRT TVs. There was a need for a larger screen, but people have this habit of sticking with something tried and tested. A revolution in thought and willingness for greater expenditure needs a product revolution as well, and that's exactly what large screen, flat panel displays have caused — a revolution. In more ways than one.

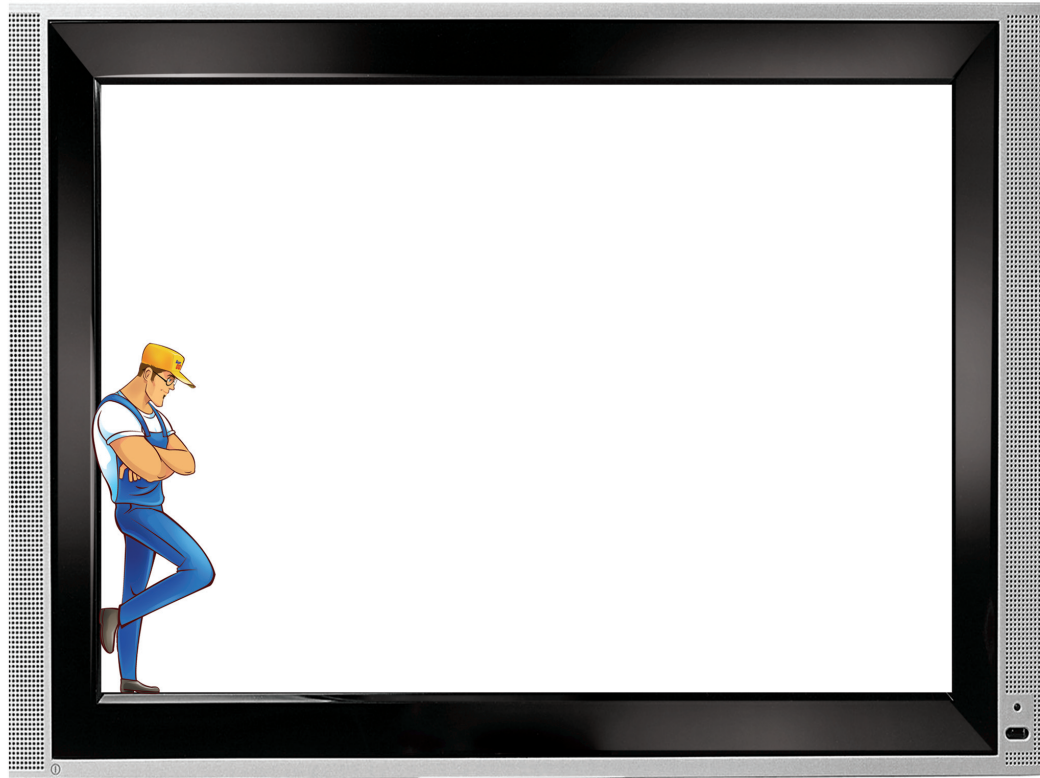
Of course it wasn't their revolution originally. The whole visual realism and immersive entertainment revolution has been cooking for some time now. NVIDIA and ATI have done their part with respect to PC and console multimedia and gaming. Then the whole HD resolution happened around two years back in India. Till this day I still find many people chanting the phrase HD like a mantra without knowing what

720p means. Now 1080p and Blu-Ray are technologies that have actually arrived, and are not just sitting somewhere over the horizon. Whichever way you look at it, i.e. PC, console, television or home entertainment system, a regular CRT TV just doesn't cut the mustard for most users out there. Large screen plasma and LCD displays are here, and they are surprisingly affordable, a factor which has led to high rates of adoption. Falling manufacturing costs are a result of optimised manufacturing procedures as well as burgeoning demand. These binaries mean that flat screen panels are one of the sizzling categories of this decade.

Most people who have the means are looking to go flat. But ask them about what they want and they're clueless, other than a few isolated thoughts like "Oh, I want a large screen for a home entertainment system", or "I want to watch HD movies, and my CRT doesn't support HD".

First, let me dispel some myths. HD is a largely abused term. It's a standard for video resolution, not a guarantee of quality. There are compatibility issues involved too. For example, if you have an HD-ready LCD TV but your DVD player doesn't support HD, you do not benefit at all. If your player also supports HD resolutions, but you are using this setup to watch DVD movies, there is no use — you aren't even experiencing half of what the hardware you've paid for is worth. Also the term "high definition" is very ambiguous — a TV supporting a resolution of 720 x 480 pixels is also a high definition TV, but it doesn't support 720p — which is a common resolution for most HD movies. So don't just go out there and ask for an HD-ready display, or at least read this before you do!

As a prospective flat screen consumer you will also have a dilemma of which panel technology to choose, i.e. LCD or plasma, and what screen size to pick. Then there are a host of other features like contrast ratios, brightness levels and resolutions to look at. There is also a sweet spot in terms of price for flat panels, and finding and exploiting this sweet spot is up to you. For example, the difference in price between a 32-inch and 42-inch LCD TV may be Rs 70,000, while the difference between a 26-inch and a 32-inch may be just Rs 6,000. Obviously, if you want a flat panel, in this case, a 32-inch makes the most sense. In general, LCD panels are available in smaller sizes from 22 inches all the way to 80 inches. Plasma panels on the other hand start from a screen size of 42 inches and go all the way



up to a colossal 110 inches, although the larger the panel, the costlier it is, and the really large panels can be astronomically priced. LCDs are generally good value for money up to the 32-inch size bracket, after which 37- and 42-inch categories become disproportionately costly. However, Plasmas are even more expensive, and if you want a 42-inch panel, a plasma TV would cost more than an LCD. But as the screen size goes up (above 42 inches) LCDs cost prices rise drastically. Therefore a 65-inch plasma TV will cost less than an LCD display of the same size.

There are some misnomers related to the two technologies. People believe that plasma displays have short life spans, and their visual quality degrades over time. While this was true some years ago, remember that plasma displays are still a developing technology and very new, so advancements are constantly being made. The plasmas of today have around the same lifespan as LCD panels which range broadly from 27,000 hours to 65,000 hours. This depends on the panel type and its rated life.

In terms of the quality between plasma and LCD, plasma displays tend to have a slightly better contrast ratio and much better colour. While an LCD's colour may be very bright and lively, and this is often mistaken for crispness, plasmas have more natural colours. Plasma displays on the whole are superior in comparison to LCD displays, although bear in mind that within both categories quality also differs, and this is dependent on the panel type used.

There is also the small issue of the relation between screen size and viewing distance. Each screen has a certain native resolution which is the total number of pixels that can be displayed. For example, the native resolution of 32- and 42-inch panels may be the same. What happens when viewing both from the same distance, slight pixelation will be visible on the larger screen, since there is a larger gap between adjacent pixels, the distance is called the pixel pitch (the same concept in CRTs is referred to as dot pitch).

When looking at the specification contrast ratio for each panel remember to look at the static contrast ratio, (SCR), and not the dynamic contrast ratio, (DCR); this figure is misleading,

as it measures the maximum and minimum intensity possible over a short period of time, and not instantaneously. For example a DCR of 20000:1 is totally unrealistic — bear in mind that this figure is hardly possible — and a panel with such a contrast ratio would be prohibitively expensive. Even an SCR of 5000:1 is hardly likely. The best panels would have a true SCR in the region of 3,000:1. In fact, we cannot notice anything more than 1,500:1 in most circumstances. It is also possible to use tests like DisplayMate to judge the contrast ratio, brightness, and colour rendition of a panel before you purchase it, and being a small application (less than 1 MB in size) you can easily carry it on a CD or thumb drive. Of course, not all vendors will let you test the TVs on display, although I was able to do so at a particular mall. There's no harm in trying, is there? Please do not take manufacturers' quoted figures as gospel truth — most of them overstate such parameters.

I visited a number of showrooms that stock large screen displays, including malls and chain stores like Next, Croma and Reliance Digital. In general 32-inch LCDs are hot right now. Their price tends to be just about Rs 8,000 to Rs 12,000 more than 26-inch displays and people who want to watch HD and Blu-Ray content flock for a larger display. Anything below 26 inches is not hot at the moment — especially considering LCD monitors of the 22- and 24-inch size categories are readily available. 37-inch LCDs are available for at a broad price range of Rs 65,000 to Rs 85,000. The 42-inch size category is priced at Rs 95,000 and above. Some of the Samsung Bordeaux and LG Scarlet models are really exorbitantly priced. I came across a new Samsung 52-inch Bordeaux A1 series LCD priced at a whopping Rs 2,20,000. Amongst the 32-inch models there is an interesting divide. Some LCD displays sport a native resolution of 1366 x 768 pixels, so they can display content up to 720p natively. Some 32-inch models, especially those from LG and Samsung, support 1920 x 1080 pixels natively — therefore 1080p is possible in all its native glory. Normally, native 1080p is a resolution reserved for only 42-inch (and above) TVs. Obviously a display with native 1080p support is costlier than a regular 720p display — and I found this difference within brands to be around

Rs 20,000 on average. Samsung's Bordeaux and LG's Scarlet series are the hottest of the lot. They're both great lookers — the new Bordeaux has piano black finish with a wine red streak of red running around the bezel — and this colour isn't painted on to the finish; it's merged into the body material, and looks ultra sexy. The Scarlet series has a beautiful stand and body finish; these TVs are a real work of art. I was surprised at seeing very few Sony Bravia LCDs — according to most of the attendants LG and Samsung are more popular, and Sony's are perceived as being costlier, although in reality this isn't always so. Also available was Vu, a UK based brand that's being marketed in India by Zenith. Its 32-inch TV was well priced — Rs 40,000, which is around Rs 18,000 cheaper than Samsung's 32-inch Bordeaux and Rs 23,000 cheaper than LG's 32-inch Scarlet model, which are the costliest 32-inch TVs on display. Samsung's 26-inch Mosel LCD TV is available for Rs 31,000, and the 32-inch version is priced at Rs 41,000 — much better than the Bordeaux — but these panels have lower specifications, not to mention the finish and body shape are not as attractive.

I came across a Hitachi 55-inch plasma panel belong to their 01 series. This was priced at Rs 95,000 and seemed to have very good contrast and the colours were absolutely superb. A very good buy, if you're shopping for a really large display. The stand was rather sleek although its simple looking and lacks the stylish contouring many look for. The bezel is slim and this makes up for the plain Jane stand.

For me, a 32-inch LCD panel makes the most sense. I'd go with a 1080p supporting panel. Do remember that most chain electronics stores like Croma, Reliance Digital and Next have monthly schemes going. You can actually purchase such a display on instalments, thus reducing the load of a large one time payment. 32 inches is also the magic figure if you plan on hooking up an Xbox 360 or a PS3 to your display, anything larger means you should be sitting further away than most gamers do. If you have a really large living room, and want something bigger, I recommend a 42-inch LCD, or a 55-inch plasma. The prices of 52-inch LCD displays are too high to even think about them at the moment.